

Total marks 36**Attempt questions 1-3****Answer each question in a SEPARATE writing booklet.**

- Question 1 (12 Marks) Use a SEPARATE writing booklet** **Marks**
- (a) If the velocity of a particle is given by $v = \frac{1}{2t+1} \text{ ms}^{-1}$ and it is initially at $x = 1$, find an expression for the displacement x in terms of t . **2**
- (b) A particle undergoes Simple Harmonic Motion about the origin O. Its displacement, x centimetres from the origin at time t seconds, is given by $x = 4 \cos\left(2t + \frac{\pi}{3}\right)$
- (i) Express the acceleration as a function of displacement. **1**
- (ii) Write down the amplitude of the motion. **1**
- (iii) Find the maximum speed and the time at which it first occurs. **2**
- (c) The acceleration of an object moving along the x -axis is given by $a = 10x - 5x^3 \text{ cm/s}^2$. It is released from rest at $x = 2$.
- (i) If v is the velocity, show that $v^2 = x^2(10 - 2.5x^2)$. **3**
- (ii) In which direction does the object first move? Show why. **2**
- (iii) Where does the object next come to rest? **1**

Question 2 12 Marks Use a SEPARATE writing booklet Marks

- (a) A baseball pitcher throws his fastball ball at 40 metres / sec . He releases the ball 1.8 metres above the ground at an angle of 2° above the horizontal and at a distance of 18.4 metres from the batter.

The equations of motion for the ball are: $\ddot{x} = 0$ and $\ddot{y} = -10$.

Take the origin to be the ground directly below the point where the pitcher releases the ball.

- (i) Using calculus, prove that the coordinates of the ball at time t are given by:
 $x = 40t \cos(2^\circ)$ and $y = -5t^2 + 40t \sin(2^\circ) + 1.8$ **3**
- (ii) Find the time it takes for the ball to reach the batter. **1**
- (iii) To be a “strike” for a particular batter, the ball must pass the batter between 53cm and 115cm above the ground. Was this ball a strike? Show your reasoning clearly. **2**
- (iv) Find the angle (to the nearest minute) at which the ball arrives at the batter. **2**
- (b) Given the equations of motion of an cannonball fired from the origin with initial speed $V \text{ m / s}$ at an angle of α° are given by:
- $$x = Vt \cos \alpha \text{ and } y = -\frac{1}{2}gt^2 + Vt \sin \alpha,$$
- (i) Show that the Cartesian equation of the path of the object are given by $y = -\frac{gx^2}{2V^2} \sec^2 \alpha + x \tan \alpha$ **2**
- (ii) Show that the range on level ground is given by $\frac{V^2}{g} \sin 2\alpha$. **1**
- (iii) If $V = 33 \text{ m / s}$ and taking $g = 10 \text{ m / s}^2$, find the maximum distance that the cannonball can be fired. **1**

Question 3 (12 Marks) Use a SEPARATE writing booklet **Marks**

(a) Find the coefficient of x^6 in the expansion of $(1 - 2x^2)^5$ **2**

(b) Using the fact that $(1+x)^4(1+x)^9 = (1+x)^{13}$, show that **2**

$${}^4C_0 {}^9C_4 + {}^4C_1 {}^9C_3 + {}^4C_2 {}^9C_2 + {}^4C_3 {}^9C_1 + {}^4C_4 {}^9C_0 = {}^{13}C_4$$

(c) Let $(2 + 5x)^{20} = \sum_{k=0}^{20} T_k$ where T_k is the term in x^k .

(i) Show that $\frac{T_{k+1}}{T_k} = \frac{100 - 5k}{2k + 2} x$ **2**

(ii) If $x = \frac{1}{3}$, find the greatest term in factored form. **3**

(d) (i) Use the binomial theorem to obtain an expansion for $(1+x)^{2n} + (1-x)^{2n}$ where n is a positive integer. **1**

(ii) Hence evaluate $1 + {}^{20}C_2 + {}^{20}C_4 + \dots + {}^{20}C_{20}$. **2**

End of Paper

Question 1.

(a) $v = \frac{1}{2t+1}$ $t=0, x=1$.

$$x = \frac{1}{2} \int \frac{2}{2t+1} dt$$

$$x = \frac{1}{2} \ln(2t+1) + C$$

when $t=0, x=1$

$$1 = \frac{1}{2} \ln 1 + C$$

$$C = 1$$

$$\therefore x = \frac{1}{2} \ln(2t+1) + 1$$

(b) $x = 4 \cos(2t + \pi/3)$

$$\dot{x} = -8 \sin(2t + \pi/3)$$

$$\ddot{x} = -16 \cos(2t + \pi/3)$$

(i) $\ddot{x} = -4x$

(ii) amp is 4

(iii) max speed is where $\sin(2t + \pi/3) = 1$

and is 8 cm/sec

which is when $2t + \pi/3 = \pi/2$

$$2t = \pi/6$$

$$t = \pi/12 \text{ sec}$$

(c) $a = 10x - 5x^3$ $t=0, v=0, x=2$

(i) $\frac{d^2 v^2}{dx^2} = 10x - 5x^3$

Integrating w.r.t x .

$$\frac{1}{2} v^2 = 5x^2 - \frac{5}{4} x^4 + C$$

when $v=0, x=2$

$$0 = 20 - 20 + C \therefore C=0$$

$$\therefore \frac{1}{2} v^2 = 5x^2 - \frac{5}{4} x^4$$

$$v^2 = 10x^2 - \frac{5}{2} x^4$$

(ii) since initial velocity is zero, direction is determined by acceleration at $t=0$ $a = 10 \times 2 - 5 \times 2^3$

$$= -20 \text{ cm/s}^2$$

$\therefore$ object will move to the left

(iii) $v^2 = x^2(10 - \frac{5}{2} x^2)$

moving left from $x=2$, next stationary point is $x=0$

If just integrate w.r.t x
not using $\frac{d^2 v^2}{dx^2}$ then no
marks.

Needs to show clearly
why it moves to
the left.

Q2

a) i) $\ddot{x} = 0$
 $\dot{x} = \int dt$

$= C$
 $t=0 \quad \dot{x} = 40 \cos 2^\circ$

$x = \int 40 \cos 2^\circ dt$
 $= 40t \cos 2^\circ + C$

$t=0 \quad x=0$

$x = 40t \cos 2^\circ \checkmark$

$\ddot{y} = -10$

$\dot{y} = \int -10 dt$

$\dot{y} = -10t + C$

$t=0 \quad \dot{y} = 40 \sin 2^\circ$

$\dot{y} = -10t + 40 \sin 2^\circ \checkmark$

$y = \int -10t + 40 \sin 2^\circ dt$
 $= -5t^2 + 40t \sin 2^\circ + C$

$t=0 \quad y = 1.8$

$y = -5t^2 + 40t \sin 2^\circ + 1.8 \checkmark$

ii) $x = 18.4$

$18.4 = 40t \cos 2^\circ$

$t = \frac{18.4}{40 \cos 2^\circ}$

$\approx 0.4602803904 \text{ sec}$

iii) $t = 0.4602803904 \checkmark$

$y = -5(0.46)^2 + 40(0.46) \sin 2^\circ + 1.8 \checkmark$

$= 1.38 \text{ m}$

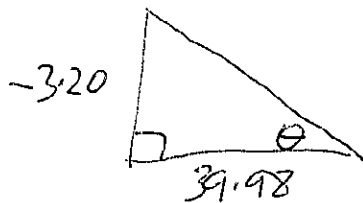
Since it is greater than 115cm it is not a strike

iv) $t = 0.46$

$\dot{x} = 40 \cos 2^\circ = 39.98 \checkmark$

$\dot{y} = -10(0.46) + 40 \sin 2^\circ$

$= -3.20$



$\tan \theta = \frac{3.20}{39.98} \checkmark$

$\theta = 4^\circ 35'$

~~Speed = $\sqrt{(-3.20)^2 + (39.98)^2}$
 $\approx 40.1 \text{ m/s}$~~

b) i) $x = Vt \cos \alpha$ ①

$y = -\frac{1}{2}gt^2 + Vt \sin \alpha$ ②

From ① $t = \frac{x}{V \cos \alpha}$ sub in ② ✓

$y = -\frac{1}{2}g \left(\frac{x}{V \cos \alpha} \right)^2 + V \left(\frac{x}{V \cos \alpha} \right) \sin \alpha$ ✓

$= -\frac{1}{2}g \frac{x^2}{V^2} \sec^2 \alpha + x \tan \alpha$

ii) Range $y = 0$

~~or~~ $0 = -\frac{1}{2}g \frac{x^2}{V^2} \sec^2 \alpha + x \tan \alpha$

$y = 0$ or $-\frac{gx}{2V^2} \sec^2 \alpha + \tan \alpha = 0$

$x = \frac{2V^2 \tan \alpha}{g \sec^2 \alpha}$

$= \frac{V^2}{g} 2 \cos^2 \alpha \tan \alpha$ ✓

$= \frac{V^2}{g} 2 \sin \alpha \cos \alpha$

$= \frac{V^2}{g} \sin 2\alpha$

iii) Range $= \frac{33^2}{10} \times 1$ Max when $\alpha = \frac{\pi}{4}$
 $= 108.9 \text{ m}$ ✓

Question 3.

$$(a) (1-2x^2)^5 = \sum_{i=0}^5 \boxed{{}^5C_i (-2x^2)^i} \checkmark$$

$$= \sum_{i=0}^5 {}^5C_i (-2)^i x^{2i}$$

For term in x^6 $2i=6$ i.e. $i=3$.

$$\therefore \text{coeff of } x^6 = {}^5C_3 (-2)^3$$

$$= -80 \checkmark$$

$$(d) (1+x)^{2n} + (1-x)^{2n} = \left({}^{2n}C_0 + {}^{2n}C_1 x + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_{2n} x^{2n} \right)$$

$$+ \left({}^{2n}C_0 - {}^{2n}C_1 x + {}^{2n}C_2 x^2 - \dots + {}^{2n}C_{2n} x^{2n} \right)$$

$$(1+x)^{2n} + (1-x)^{2n} = 2 \left({}^{2n}C_0 + {}^{2n}C_2 x^2 + \dots + {}^{2n}C_{2n} x^{2n} \right) \checkmark$$

sub $x=1$ in both side and $n=10$.

$$(1+1)^{20} + 0 = 2 \left({}^{20}C_0 + {}^{20}C_2 + \dots + {}^{20}C_{20} \right) \checkmark$$

$$2^{20} = 2 \left(1 + {}^{20}C_2 + \dots + {}^{20}C_{20} \right)$$

$$\therefore 1 + {}^{20}C_2 + {}^{20}C_4 + \dots + {}^{20}C_{20} = 2^{19} \checkmark$$

$$(e) \text{ LHS} = ({}^4C_0 + {}^4C_1 x + {}^4C_2 x^2 + {}^4C_3 x^3 + {}^4C_4 x^4) ({}^9C_0 + {}^9C_1 x + {}^9C_2 x^2 + {}^9C_3 x^3 + {}^9C_4 x^4 \dots) \checkmark$$

Note ${}^{13}C_4$ is coeff of x^4 in expansion of $(1+x)^{13}$ ✓

$\therefore$ find coeff of x^4 in expansion of $(1+x)^4 (1+x)^9$

$$i.e. {}^4C_0 {}^9C_4 + {}^4C_1 {}^9C_3 + {}^4C_2 {}^9C_2 + {}^4C_3 {}^9C_1 + {}^4C_4 {}^9C_0$$

Equating coefficients give result.

$$(f)(i) \text{ In } (2+5x)^{20}, T_k = {}^{20}C_k 2^{20-k} (5x)^k \checkmark$$

$$= {}^{20}C_k 2^{20-k} 5^k x^k$$

$$T_{k+1} = {}^{20}C_{k+1} 2^{19-k} 5^{k+1} x^{k+1}$$

$$\frac{T_{k+1}}{T_k} = \frac{{}^{20}C_{k+1} 2^{19-k} 5^{k+1} x^{k+1}}{{}^{20}C_k 2^{20-k} 5^k x^k} \checkmark$$

$$= \frac{5(20-k)}{2(k+1)} \times x$$

$$= \frac{(100-5k)}{2k+2} \times x$$

$$\frac{{}^{20}C_{k+1}}{{}^{20}C_k} = \frac{20!}{(k+1)!(20-(k+1))!} \times \frac{k!(20-k)!}{20!}$$

$$= \frac{(20-k)}{(k+1)} \checkmark$$

(10) If $x = \frac{1}{3}$ $\frac{T_{k+1}}{T_k} = \frac{100-5k}{6k+6}$

For greatest term find k such that

$$\frac{T_{k+1}}{T_k} > 1$$

i.e. $\frac{100-5k}{6k+6} > 1$ ✓

i.e. $100-5k > 6k+6$

$$11k < 94$$

$$k < 8\frac{6}{11}$$
 ✓

i.e. $k=8$ is highest value for which this is true

$\therefore T_9 > T_8 > T_7$ etc. i.e. T_9 is greatest

$$T_9 = {}^{20}C_9 \cdot 2^{11} \cdot 5^9 \cdot \left(\frac{1}{3}\right)^9$$
 ✓